# Terahertz Wireless Networks: Opportunities, Challenges, and Enabling Technologies

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***Abstract*—Terahertz (THz) wireless communication offers large bandwidth and high data rates, but its performance is affected by propagation loss, physical blockage, and beam misalignment. This project investigates the problem of signal blockage in indoor THz communication and evaluates reflection-based beam steering as a possible solution.**

**The study combines a literature review with practical simulation. An open-source MATLAB implementation was used to examine THz channel propagation and molecular absorption. In addition, a Python-based 3D ray-tracing simulation was implemented using NVIDIA Sionna RT at 300 GHz.**

**The simulation evaluated clear line-of-sight (LoS), blocked LoS, specular reflection, and directional beam steering.**

**When the direct communication path was blocked, an alternative reflected path was identified. Beam sweeping was performed from $-90^\circ$ to $90^\circ$ in steps of $5^\circ$.**

**The best tested orientation was $50^\circ$, reducing effective link loss from approximately 102.31 dB to 95.42 dB, an improvement of approximately 6.88 dB.**

**Under the assumed link-budget conditions, the theoretical Shannon capacity increased from 41.90 Mbps to 193.83 Mbps.**

**The results demonstrate that reflection and beam steering can improve link performance in the evaluated static indoor scenario. However, real-time beam tracking and network-level reliability were not evaluated.**

***Index Terms*—Terahertz communications, wireless networks, beamforming, channel modeling, network reliability.**

## I. Introduction

The demand for high-data-rate wireless services motivates research into frequency bands beyond those commonly used in existing networks. Terahertz (THz) communication is being investigated as one possible approach because of the wide bandwidths available in selected frequency windows [1].

However, wide bandwidth alone does not guarantee a reliable wireless connection. THz network design must account for propagation loss, frequency-dependent molecular absorption, blockage, and hardware limitations. Channel measurements and appropriate models are therefore important for understanding practical deployment conditions [2].

Directional antennas and beamforming can help establish useful communication links. However, narrow beams require alignment between transmitters and receivers. User movement and obstacles can make link maintenance more difficult. These challenges connect physical-layer design with network tasks such as beam management and maintaining connectivity [3].

This article presents a focused narrative review of THz wireless networks. It addresses the following question: How can THz networks benefit from large bandwidth while maintaining reliable connections under propagation, blockage, and beam-alignment constraints?

**Problem Statement and Proposed Solution.**

The main problem addressed in this project is signal blockage in indoor Terahertz (THz) wireless communication. When an obstacle blocks the direct Line-of-Sight (LoS) path between the transmitter and receiver, the communication link may become unavailable.

To investigate a possible solution, this project uses 3D ray-tracing simulation at 300 GHz to find an alternative reflected signal path. Directional beam steering is then applied to identify a transmitter orientation that reduces signal loss. The proposed approach is evaluated by comparing link performance before and after beam steering.

The review draws on three broad surveys [1]–[3] and examines two original studies on indoor channel modeling and adaptive beam tracking [4], [5]. These studies illustrate complementary approaches to understanding and maintaining THz links. The discussion compares their methods, evidence, and limitations, with particular attention to network reliability.

The literature review is selective rather than exhaustive. In addition, this study presents a 300 GHz indoor ray-tracing simulation using NVIDIA Sionna RT to evaluate line-of-sight propagation, blockage, specular reflection, and directional beam steering.

The review first introduces THz fundamentals and potential applications. It then examines propagation and networking challenges, followed by enabling technologies and a comparison of selected studies. Finally, it discusses design trade-offs, open research directions, and the main conclusions.

## II. THz Network Fundamentals

Terahertz (THz) wireless communication generally refers to the use of frequencies between 0.1 and 10 THz [2]. This frequency range offers substantial bandwidth that can support high-capacity wireless links and emerging applications [1]. However, the performance of THz networks depends not only on the available bandwidth but also on the propagation environment, transmission distance, and hardware capabilities.

### A. Related Work

Previous studies have examined THz wireless communication from different perspectives. Shafie et al. [1] reviewed major challenges and technological advancements across the physical, link, and network layers, emphasizing spectrum management, antenna technologies, and beamforming.

Han et al. [2] focused on THz channel measurements, propagation characteristics, and channel modeling techniques. Their survey provides an important foundation for understanding the behavior of THz wireless links.

Jiang et al. [3] presented a broader review of THz communications and sensing, covering propagation, transceiver technologies, network architectures, and experimental developments.

Building on these studies, the present review focuses on the relationship between THz propagation characteristics, enabling technologies, and network reliability. Particular attention is given to the trade-offs between communication performance, connection stability, and implementation complexity.

### B. Propagation Characteristics

THz signals experience free-space propagation loss and frequency-dependent molecular absorption [2]. In particular, atmospheric molecules, including water vapor, absorb electromagnetic energy at specific frequencies. This creates transmission windows in which signal attenuation is relatively low compared with neighboring frequencies.

For a line-of-sight (LoS) THz link, the combined free-space spreading and molecular absorption loss can be expressed as

$$L(f,d) = \left(\frac{4\pi f d}{c}\right)^2 e^{\kappa(f)d}, \tag{1}$$

where $f$ is the operating frequency in hertz, $d$ is the transmission distance in meters, $c$ is the speed of light in meters per second, and $\kappa(f)$ is the power absorption coefficient in inverse meters. Here, $L(f,d)$ is a dimensionless power-loss factor in linear scale, not in decibels. The expression combines free-space spreading and molecular absorption for a line-of-sight path; antenna gains and additional losses due to blockage or misalignment are not included [2].

The severity of these effects depends on operating frequency, communication distance, and environmental conditions. Therefore, accurate channel modeling is essential for evaluating the feasibility and reliability of THz wireless links.

Chen et al. [4] conducted channel measurements between 130 and 143 GHz in a typical meeting room using directional antennas. The study examined line-of-sight path loss and the temporal and spatial characteristics of multipath propagation. A hybrid model combining ray tracing and statistical methods was developed and evaluated against the measurements. The model showed better agreement with the measured temporal-spatial characteristics than conventional statistical and geometry-based stochastic models.

For this review, the main implication is that realistic propagation models provide a stronger basis for evaluating THz links than free-space loss calculations alone. However, the reported validation concerns a particular indoor environment and frequency range. Its applicability to other deployment conditions therefore requires further validation. Accurate channel modeling should also be distinguished from network reliability evaluation, which must additionally consider mobility, blockage, and link recovery.

### C. Directional Communication

To improve the link budget, THz communication systems commonly rely on high-gain directional antennas and beamforming techniques [1]. These technologies concentrate transmitted energy toward the intended receiver, improving the received signal strength.

However, narrow directional beams require accurate alignment between the transmitter and receiver. Mobility, environmental obstacles, and sudden blockage can interrupt communication and require beam reconfiguration.

As illustrated in Fig. 1, physical blockage can interrupt a directional line-of-sight THz communication link.

Consequently, reliable THz network operation requires coordination between physical-layer communication techniques and network-level mechanisms such as beam tracking, link recovery, and connectivity management [3].

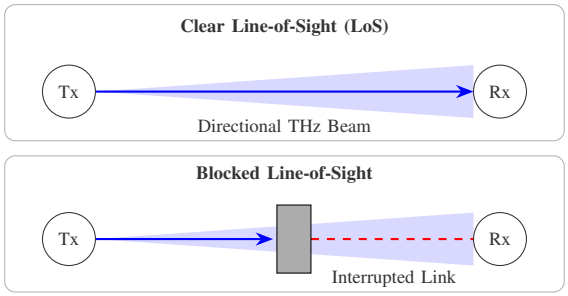



Fig. 1: Conceptual illustration of directional THz communication under clear and blocked line-of-sight conditions.

### D. Opportunities and Applications

The availability of large bandwidths in the THz frequency range creates opportunities for extremely high-data-rate wireless communication. Compared with conventional lower-frequency systems, THz technologies can potentially support substantially higher transmission capacities, making them attractive for bandwidth-intensive applications [1].

For an ideal additive white Gaussian noise (AWGN) channel, the theoretical channel capacity is described by Shannon's formula:

$$C = B\log_2(1+\mathrm{SNR}), \tag{2}$$

where $C$ is the theoretical channel capacity in bits per second, $B$ is the available bandwidth in hertz, and SNR is the signal-to-noise ratio in linear scale. This equation illustrates that increasing the usable bandwidth can improve the achievable data rate when the SNR is maintained. However, practical THz communication performance is also limited by propagation loss, hardware constraints, and link reliability [1].

One promising application is short-range, high-capacity indoor communication, where THz links may support rapid

data transfer between nearby devices. THz communication is also being investigated for wireless backhaul and fronthaul connections, which can help transport large amounts of data between network infrastructure components. Such applications could reduce reliance on wired connections in suitable deployment scenarios [1].

Another important opportunity is the integration of wireless communication and sensing. The high operating frequencies and large bandwidths of THz systems can support high-resolution environmental sensing and precise localization. Combining communication and sensing capabilities may enable new applications in intelligent transportation, industrial automation, and future wireless networks [3].

Despite these opportunities, their practical realization depends on overcoming propagation, hardware, and networking limitations. Therefore, the potential benefits of THz communication must be evaluated together with the reliability challenges discussed in the following section.

## III. Challenges in THz Wireless Networks

While the previous section introduced the fundamental characteristics of THz communication, deploying reliable THz wireless networks presents several practical challenges. These include propagation attenuation, signal blockage, beam misalignment, and hardware limitations [1].

These challenges affect communication range, link stability, and overall network performance. Understanding their impact is essential for evaluating the technologies and design approaches discussed in the following sections.

### *A. Propagation Loss and Molecular Absorption*

One of the major challenges in THz wireless networks is the significant attenuation experienced by signals during propagation. For fixed antenna gains, free-space path loss increases with operating frequency, making the communication link more sensitive to transmission distance. This can reduce the received signal power and limit the achievable communication range [2].

Another important limitation is molecular absorption, which introduces frequency-dependent attenuation. Atmospheric molecules, particularly water vapor, absorb energy at specific THz frequencies. As a result, the available transmission windows and achievable link distances depend on the selected frequency and environmental conditions [2].

From a network design perspective, these propagation characteristics create important trade-offs between bandwidth, coverage, and reliability. Although THz communication offers opportunities for extremely high data rates, practical systems must carefully select operating frequencies, transmission distances, and antenna configurations. Directional beamforming and suitable network deployment strategies can help address these limitations, but they also introduce additional design complexity [1].

### *B. Blockage and Mobility*

Blockage represents a major challenge in THz wireless networks because communication often relies on highly directional line-of-sight (LoS) links. Physical obstacles, such as walls, furniture, and the human body, can significantly attenuate or interrupt the transmitted signal. Although reflected propagation paths may provide alternative connections, these paths can experience additional attenuation and may not always support reliable communication [2].

User mobility introduces further difficulties because the relative positions of transmitters and receivers change over time. Even small movements can affect the alignment of narrow directional beams, potentially reducing the received signal strength and causing temporary link interruptions. Consequently, maintaining stable connectivity requires continuous monitoring of changing channel and network conditions [1].

From a networking perspective, blockage and mobility can increase the frequency of beam realignment, link recovery, and handover procedures. These operations introduce signaling overhead and may increase communication latency, especially in dynamic environments with moving users and obstacles. Therefore, reliable THz networks require efficient mobility management and mechanisms capable of responding rapidly to sudden changes in connectivity [3].

### *C. Beam Alignment and Tracking*

Beam alignment is a critical challenge in THz wireless networks because communication systems typically employ highly directional antennas to achieve sufficient received signal strength. Establishing a reliable connection requires accurate alignment between the transmitter and receiver beams. Misalignment can significantly reduce antenna gain and degrade communication performance [1].

Maintaining beam alignment becomes more difficult in dynamic environments where users or devices change their positions or orientations. Beam tracking mechanisms are therefore required to adjust the transmission direction and maintain connectivity.

However, frequent beam training and realignment procedures introduce additional signaling overhead, energy consumption, and communication delays [3].

An important design trade-off exists between beamwidth and alignment complexity. Narrow beams provide higher directional antenna gain but are more sensitive to positioning errors and mobility. Wider beams simplify alignment and improve tolerance to movement, but generally provide lower directional gain. Consequently, efficient beam management strategies must balance link quality, tracking overhead, and network reliability [1].

### *D. Hardware and Energy Constraints*

The practical implementation of THz wireless networks faces significant hardware challenges due to the extremely high operating frequencies. Generating, transmitting, and detecting THz signals require specialized electronic and photonic components. Current hardware technologies face limitations related to output power, receiver sensitivity, conversion efficiency, and phase noise. These limitations can reduce communication range and affect overall link performance [1].

Energy consumption is another important concern, particularly in systems employing large antenna arrays and advanced beamforming architectures. Signal processing, beam training, and the operation of multiple radio-frequency components can increase power requirements. This creates challenges for mobile devices and energy-constrained network nodes, where battery capacity and thermal management must be considered [3].

From a network design perspective, hardware limitations influence coverage, deployment cost, and energy efficiency. Increasing transmission power or using more complex antenna architectures may improve link performance, but these approaches can also increase hardware complexity and power consumption. Therefore, practical THz networks must balance communication capacity, reliability, hardware feasibility, and energy efficiency [1].

### *E. Network Reliability*

Network reliability is a critical concern in THz wireless communication because maintaining stable connectivity depends on several interconnected factors. Propagation attenuation, molecular absorption, physical blockage, and beam misalignment can significantly affect link availability and communication quality. These challenges become particularly important in dynamic environments where channel conditions change rapidly [1].

Unlike conventional lower-frequency wireless systems, THz networks often depend on highly directional links that can be vulnerable to sudden interruptions. Link failures may cause packet losses, increased latency, and reductions in effective throughput. Therefore, evaluating network performance requires consideration of reliability indicators such as link availability, packet delivery performance, and recovery delay [3].

Improving reliability requires coordinated solutions involving beam management, adaptive transmission techniques, and alternative communication paths. For example, beam realignment and switching to an available backup link may help restore connectivity after blockage. However, these mechanisms introduce additional signaling overhead, processing requirements, and possible delays. Consequently, THz network design must balance high data rates with connection stability, resource efficiency, and recovery performance [1].

## IV. Enabling Technologies for THz Wireless Networks

Addressing the challenges discussed in the previous section requires advanced communication technologies and efficient network design strategies. Several enabling approaches have been investigated to improve the capacity, coverage, and reliability of THz wireless networks. These include directional beamforming, large antenna arrays, beam management techniques, and adaptive communication mechanisms [1].

Although these technologies can mitigate propagation limitations and improve link stability, each introduces trade-offs involving hardware complexity, energy consumption, and signaling overhead. Understanding these trade-offs is essential for developing practical and reliable THz communication systems [3].

### *A. Directional Beamforming*

Directional beamforming is one of the key enabling technologies for THz wireless communication. It improves link performance by concentrating electromagnetic energy toward the intended receiver rather than distributing it equally in all directions. This directional gain helps compensate for high propagation losses and supports more reliable high-data-rate communication over suitable distances [1].

Beamforming can be implemented using analog, digital, or hybrid architectures. Analog beamforming controls antenna signals using radio-frequency phase shifters and generally requires fewer RF chains. Digital beamforming provides greater flexibility by processing signals separately across multiple RF chains, but often increases hardware complexity and power consumption. Hybrid beamforming combines analog and digital processing to balance performance, flexibility, and implementation cost [3].

Despite these advantages, directional beamforming introduces important limitations. Narrow beams require accurate alignment between the transmitter and receiver, making communication sensitive to mobility and physical blockage. Furthermore, beam training and adjustment procedures consume network resources and may introduce additional latency. Therefore, efficient beamforming design must balance directional gain, implementation complexity, and connection reliability [1].

### *B. Large Antenna Arrays and MIMO Technologies*

Large antenna arrays are important enabling technologies for THz wireless networks because the short wavelengths at THz frequencies allow many antenna elements to be integrated into relatively compact physical structures. By combining signals across multiple antenna elements, these arrays can provide high directional gain, improving the link budget and supporting high-capacity communication [1].

Multiple-Input Multiple-Output (MIMO) technology uses multiple transmitting and receiving antennas to improve wireless communication performance. In THz systems, MIMO architectures can support beamforming and, under suitable channel conditions, spatial multiplexing to increase data transmission capacity. However, the potential multiplexing gains depend on channel characteristics, antenna configuration, and the availability of sufficiently independent spatial communication paths [3].

Despite their advantages, large antenna arrays and MIMO systems introduce significant implementation challenges. Increasing the number of antenna elements may require more complex signal processing, accurate channel estimation, and efficient hardware architectures. Fully digital implementations can also consume considerable energy due to the required radio-frequency chains. Consequently, practical THz systems often consider hybrid antenna architectures to balance communication performance, hardware complexity, and energy efficiency [1].

### *C. Beam Management and Tracking*

Beam management is an essential technology for maintaining reliable communication in THz wireless networks.

Since THz systems commonly rely on highly directional transmission, efficient mechanisms are required to establish, monitor, and maintain suitable communication beams. Beam management generally involves initial beam training, beam selection, tracking, and recovery procedures [1].

Beam tracking enables transmitters and receivers to adjust their beam directions when users move or change their orientation. By updating beam configurations according to changing channel conditions, tracking mechanisms can reduce misalignment and help maintain stable connections. However, frequent beam updates may introduce signaling overhead, processing complexity, and additional energy consumption [3].

Beam recovery is particularly important when communication is interrupted by physical blockage or sudden changes in the propagation environment. In such cases, the network may attempt to establish an alternative beam, use an available reflected propagation path, or switch to another access point. The effectiveness of these approaches depends on the availability of suitable links and the time required to restore connectivity [1].

Karacora et al. [5] proposed event-based beam tracking combined with dynamic beamwidth adaptation for THz communications. Their formulation jointly considers maximizing the expected data rate and minimizing outage probability under transmit-power and overhead constraints. The approach adapts beamwidth to balance directional gain against tolerance to misalignment, while triggering channel estimation according to communication conditions. Simulation results showed improved average rate and reduced outages while maintaining low pilot overhead.

For this review, the key implication is that beam management should be evaluated using both communication performance and the resources required to maintain the link. A highly directional beam is useful only when its gain can be maintained without excessive tracking overhead or interruptions. However, the reported evidence is based on simulations rather than a deployed network. The findings should therefore be interpreted within the evaluated conditions, with experimental validation needed before generalizing the benefits to practical deployments.

Therefore, efficient beam management must balance tracking accuracy, recovery speed, and resource consumption. Reducing unnecessary beam training while maintaining responsiveness to mobility and blockage is an important design objective for reliable THz networks [3].

### *D. Adaptive Communication and Resource Allocation*

Adaptive communication techniques play an important role in improving the performance of THz wireless networks under changing channel conditions. Since THz propagation is affected by transmission distance, molecular absorption, and environmental factors, fixed transmission configurations may not provide satisfactory performance in all scenarios. Adaptive mechanisms can adjust communication parameters according to channel quality and network requirements [1].

Adaptive modulation and coding can improve transmission efficiency by selecting suitable modulation orders and coding rates based on channel conditions. Under favorable conditions, higher-order modulation may support increased data rates, while more robust configurations can help maintain communication quality when the received signal becomes weaker. However, these adjustments require reliable channel information and introduce additional control complexity [3].

Resource allocation is particularly important because THz channels exhibit frequency-dependent propagation characteristics. Different frequency bands may experience different levels of molecular absorption and attenuation. Therefore, frequency selection, bandwidth allocation, and transmit power management should consider the propagation characteristics and requirements of individual communication links [2].

Although adaptive communication and resource allocation can improve network efficiency, their effectiveness depends on accurate channel estimation and timely adaptation. Frequent parameter updates may increase signaling overhead and computational demands. Consequently, practical THz systems must balance achievable throughput, link reliability, adaptation complexity, and energy consumption [1].

Table I compares the two original studies discussed in this review, highlighting their evidence and implications for reliable THz communication.

The studies address complementary aspects of reliable THz communication. Chen et al. [4] provide measurement-based evidence for channel modeling, whereas Karacora et al. [5] evaluate adaptive beam management through simulations. Their results cannot be ranked directly because they use different objectives, methods, and performance measures. Taken together, they motivate evaluating beam-management strategies with measurement-informed channel models, while accounting for both outage performance and tracking overhead.

## V. 3D Ray-Tracing Simulation and Performance Evaluation

To complement the theoretical review, an indoor terahertz (THz) wireless communication simulation was conducted using NVIDIA Sionna RT at an operating frequency of 300 GHz.

The simulation investigated the effects of line-of-sight (LoS) propagation, physical blockage, specular reflection, and directional beam steering on wireless link performance.

The simulated environment was a rectangular room with dimensions of $7 \times 6 \times 3$ m. The transmitter (Tx) and receiver (Rx) were separated by 5 m and positioned at a height of 1 m.

Three main propagation conditions were investigated: clear LoS, blocked LoS, and blocked LoS with an available reflected path.

The simulation used illustrative wall material properties rather than experimentally measured THz material parameters. Frequency-dependent molecular absorption was included using an absorption-coefficient dataset.

### *A. Implementation Workflow and Tools*

The project was carried out in two stages.First, open-source MATLAB-based code obtained from GitHub [6] was executed on Ubuntu using a free MATLAB-compatible environment to examine THz propagation loss and molecular absorption. The

Table I: Comparison of Selected THz Studies

| Study and Setting | Approach | Main Evidence | Scope and Limitations |
|---|---|---|---|
| Chen et al. [4] Indoor meeting room; 130–143 GHz. | Channel measurements and a hybrid ray-tracing and statistical model. | Closer agreement with measured temporal-spatial characteristics than the conventional models considered. | Validation concerns the measured indoor setting and frequency range; it does not establish network-level reliability. |
| Karacora et al. [5] Simulation-based THz beam tracking. | Event-based channel estimation and dynamic beamwidth adaptation. | Improved average data rate and reduced outages with low pilot overhead in the evaluated simulations. | Simulation evidence; practical benefits require experimental validation under deployment-specific conditions. |

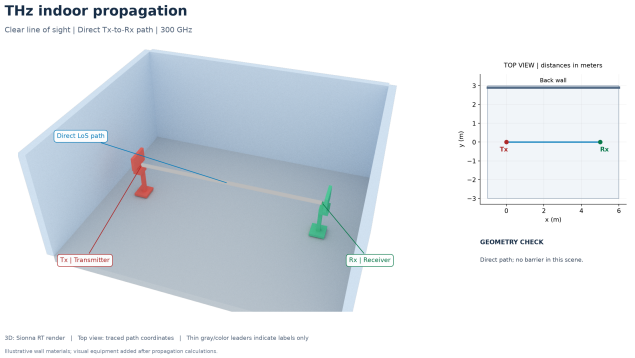



Fig. 2: Clear line-of-sight THz communication at 300 GHz using Sionna RT.

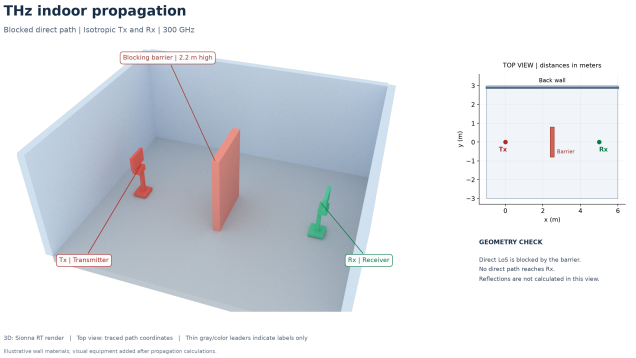



Fig. 3: Indoor THz communication scenario with a physical barrier blocking the direct LoS path.

Python scripts also load the molecular-absorption dataset and interpolate its coefficient at the operating frequency of 300 GHz.

Second, a Python-based 3D simulation was implemented using NVIDIA Sionna RT. The simulation considered clear line-of-sight (LoS), blocked LoS, a reflected propagation path, and directional beam sweeping.

The practical objective was to investigate whether a wall-reflected path, combined with beam steering, could reduce effective link loss in a static indoor scenario when the direct LoS path was blocked. This is an offline simulation-based evaluation, not a real-time beam-tracking implementation.

### *B. Clear Line-of-Sight Scenario*

Figure 2 illustrates the indoor environment under clear LoS conditions.

The red equipment represents the transmitter, while the green equipment represents the receiver. The propagation path directly connects the two devices without encountering an obstacle.

Using the directional transmitter configuration, the effective clear-LoS link loss was approximately 88.02 dB.

This result demonstrates that substantial propagation attenuation occurs even when a direct path is available, highlighting the importance of directional antennas at THz frequencies.

### *C. Blocked Line-of-Sight Scenario*

A physical barrier was introduced between the transmitter and receiver to investigate the influence of blockage.

Figure 3 presents the blocked communication scenario.

The ray-tracing simulation confirmed that the direct LoS path was unavailable in the presence of the barrier.

This result highlights the sensitivity of THz wireless links to physical obstacles.

Nevertheless, the absence of a direct path does not necessarily imply the complete loss of communication. Alternative propagation paths may still exist through reflection.

### *D. Specular Reflection and Beam Steering*

To investigate an alternative communication path, specular reflection was enabled in the ray-tracing simulation.

A valid reflected path was identified despite the blockage of the direct LoS path.

Figure 4 illustrates the resulting three-dimensional propagation.

The transmitted signal reaches the room's back wall, where a specular reflection redirects it toward the receiver.

The reflected path travels around the blocking barrier rather than passing directly through the obstacle.

With the transmitter oriented at $50^\circ$, the effective reflected-link loss was approximately 95.42 dB.

Compared with the directional clear-LoS loss of approximately 88.02 dB, the reflected configuration introduced about 7.41 dB of additional effective loss.

Both scenarios used the same directional transmitter antenna model, with an orientation suited to the corresponding propagation path.

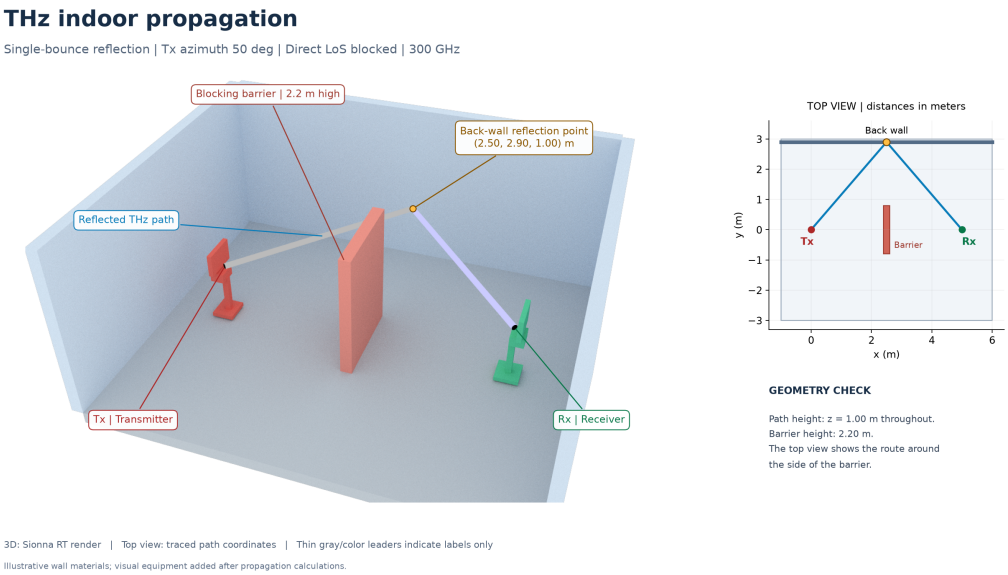



Fig. 4: Three-dimensional visualization of a blocked THz link with specular reflection from the back wall. Directional beam steering allows the transmitter to exploit the reflected propagation path.

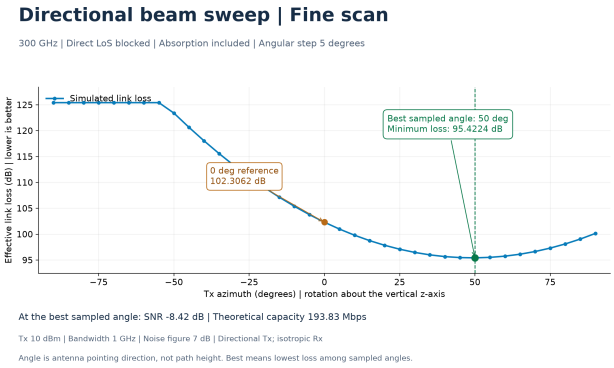



Fig. 5: Directional beam sweeping at 300 GHz. The best tested transmitter orientation is $50°$.

These results illustrate how reflection can provide an alternative wireless path when direct communication is blocked.

### *E. Directional Beam Sweeping Results*

A directional beam-sweeping experiment was performed to identify the transmitter orientation associated with the lowest effective link loss.

The transmitter used the 3GPP TR 38.901 antenna pattern, while the receiver used an isotropic antenna.

Transmit orientations between $-90°$ and $90°$ were evaluated in steps of $5°$.

Figure 5 illustrates the resulting link-loss variation.

The minimum effective link loss was approximately 95.42 dB at $50°$.

Compared with the $0°$ orientation in the same blocked environment, beam steering reduced effective link loss by approximately 6.88 dB.

Table II: Beam Steering Performance Comparison

| **Metric** | $0°$ | $50°$ |
|---|---|---|
| Loss (dB) | 102.31 | 95.42 |
| SNR (dB) | -15.31 | -8.42 |
| Capacity (Mbps) | 41.90 | 193.83 |

Under the assumed link-budget conditions, the optimized reflected link achieved an estimated SNR of $-8.42$ dB and a theoretical channel capacity of approximately 193.83 Mbps.

These findings demonstrate the importance of appropriate beam orientation when using reflected THz propagation paths.

The selected angle is the best among the tested orientations for this specific simulation geometry.

To make the contribution of beam steering explicit, Table II compares a fixed $0°$ direction with the selected $50°$ direction within the *same blocked scene*, using the same directional transmitter and isotropic receiver configurations. Beam steering reduced the effective loss by approximately 6.88 dB and increased the estimated Shannon capacity from 41.90 Mbps to 193.83 Mbps under the assumed link budget. These capacity values are theoretical estimates, not measured end-to-end throughput.

### *F. Link Budget and Performance Evaluation*

A separate link-budget evaluation was conducted using isotropic transmitter and receiver antenna patterns.

The analysis compared clear-LoS propagation with blocked propagation through a reflected path.

The assumed transmitter power was 10 dBm, the bandwidth was 1 GHz, and the receiver noise figure was 7 dB.

The theoretical capacity was evaluated using Shannon's channel-capacity formula.

Table III summarizes the calculated performance metrics.

Table III: THz Link Performance Using Isotropic Antennas

| Metric | Clear LoS | Reflection |
|---|---|---|
| Path loss (dB) | 96.02 | 103.42 |
| Rx power (dBm) | -86.02 | -93.42 |
| SNR (dB) | -9.02 | -16.42 |
| Capacity (Mbps) | 170.46 | 32.52 |

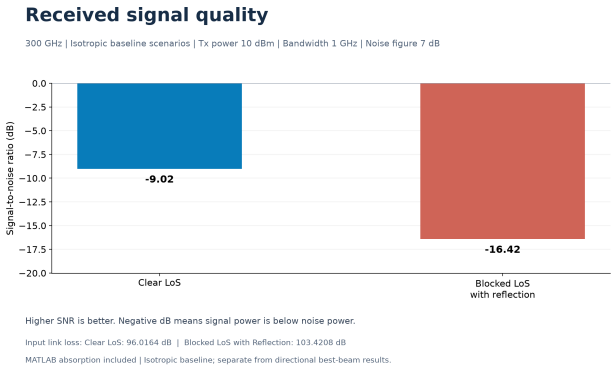



Fig. 6: SNR comparison between clear-LoS and reflected THz links using isotropic antennas.

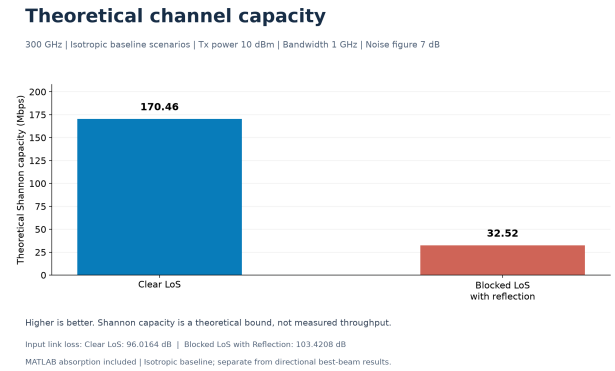



Fig. 7: Shannon theoretical channel capacity comparison between clear-LoS and reflected THz links using isotropic antennas.

*1) Signal-to-Noise Ratio:* Figure 6 compares the SNR values of the two isotropic-antenna communication scenarios.

The clear-LoS scenario achieved an SNR of approximately $-9.02$ dB.

In comparison, the blocked-reflection scenario achieved approximately $-16.42$ dB.

The reduced SNR is associated with additional propagation attenuation along the indirect reflected path.

*2) Theoretical Channel Capacity:* Figure 7 illustrates the corresponding theoretical channel-capacity results.

The theoretical capacity decreased from approximately 170.46 Mbps in the clear-LoS scenario to 32.52 Mbps in the blocked-reflection scenario.

These results indicate that the additional loss introduced by the reflected path reduces both the SNR and the theoretical channel capacity.

The reported capacity values are based on Shannon's formula and should not be interpreted as experimentally measured data throughput.

Moreover, the isotropic link-budget evaluation and directional beam-sweeping experiment represent different antenna configurations and must be interpreted separately.

### G. Discussion and Limitations

The simulation results demonstrate several important characteristics of indoor THz wireless communication.

First, direct LoS propagation provides a lower-loss communication path than the reflected alternative considered in the simulated environment.

Second, physical obstacles can eliminate the direct propagation path, making THz communication sensitive to environmental blockage.

Third, specular reflection can provide an alternative propagation route when a direct path is unavailable.

Finally, directional beam steering can improve effective link performance by aligning the transmitting antenna with an available reflected path.

However, these results are specific to the considered room geometry, antenna configurations, and illustrative material properties.

The study does not evaluate actual network throughput, mobility, beam-tracking overhead, or network-wide reliability. The optimized $50^\circ$ angle was selected from a discrete set of orientations and is specific to this room geometry and material assumptions.

Future investigations should include experimentally characterized THz materials, dynamic environments, multiple users, and adaptive beam-management algorithms.

Overall, the findings highlight the importance of realistic propagation modeling, reflected-path utilization, and directional beam control for future THz wireless networks.

## VI. Conclusions

This project investigated the challenges of terahertz (THz) wireless communication, focusing on signal blockage, propagation loss, and directional beam alignment.

The main problem addressed was the interruption of the direct line-of-sight (LoS) communication path when a physical obstacle is located between the transmitter and receiver.

To investigate this problem, an existing MATLAB-based THz channel modeling project was used to study propagation loss and molecular absorption.

A Python-based simulation was then implemented using NVIDIA Sionna RT at an operating frequency of 300 GHz.

The simulation considered a 3D indoor environment and evaluated several communication scenarios, including clear LoS, blocked LoS, specular reflection, and directional beam steering.

The proposed approach investigated the use of an alternative reflected path to maintain a potential communication route when the direct LoS path was blocked.

Directional beam sweeping was performed over transmitter orientations ranging from $-90^\circ$ to $90^\circ$ in steps of $5^\circ$.

The best tested orientation was $50^\circ$, which reduced the effective link loss from approximately 102.31 dB to 95.42 dB in the same blocked environment.

This represents an improvement of approximately 6.88 dB compared with the $0^\circ$ transmitter orientation.

Under the assumed link-budget conditions, the estimated SNR improved from $-15.31$ dB to $-8.42$ dB.

The corresponding theoretical Shannon channel capacity increased from approximately 41.90 Mbps to 193.83 Mbps.

These results demonstrate that selecting a suitable transmission direction can improve the performance of a reflected THz communication path under the evaluated conditions.

A separate isotropic-antenna analysis also confirmed that reflected propagation introduces additional path loss compared with clear LoS communication.

The simulations therefore highlight two important findings: physical blockage can eliminate the direct communication path, while reflection and directional beam steering can provide a potential alternative with improved signal quality.

However, the proposed approach was evaluated using a static simulation environment with illustrative material properties and assumed antenna patterns.

The reported channel capacities are theoretical estimates rather than experimentally measured throughput.

Furthermore, the simulation does not implement real-time beam tracking, user mobility, or automatic link recovery.

Future work could extend the simulation by introducing moving obstacles, adaptive beam selection, and more realistic THz channel measurements.

Overall, the project demonstrates how 3D ray tracing and directional beam steering can be used to investigate and partially mitigate signal blockage in indoor THz wireless communication.

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